

Connect and Protect: Networks and Network Security

Module 1

Network architecture

Get started with the course

Introduction to Course 3

Video • 1 min

Course 3 overview

Reading • 4 min

Helpful resources and tips

Reading • 4 min

Introduction to networks

Welcome to module 1

Video • 29 sec

Chris: My path to cybersecurity

Video • 3 min

What are networks?

Video • 1 min

Tina: Working in network security

Video • 2 min

Emmanuel: Useful skills for network security

Video • 2 min

Network tools

Video • 2 min

Network components, devices, and diagrams

Reading • 4 min

Practice: Design a local area network

Ungraded Plugin • 10 min

Cloud networks

Video • 2 min

Cloud computing and software-defined networks

Reading • 8 min

Identify: Functions of network tools

Ungraded Plugin • 10 min

Test your knowledge: Introduction to networks

Practice Assignment • Grade: 100%

Network communication

Introduction to network communication

Video • 2 min

The TCP/IP model

Video • 2 min

The four layers of the TCP/IP model

Video • 1 min

Learn more about the TCP/IP model

Reading • 4 min

The OSI model

Reading • 8 min

Test your knowledge: Network communication

Practice Assignment • Grade: 100%

Local and wide network communication

IP addresses and network communication

Video • 2 min

Components of network layer communication

Reading • 4 min

Test your knowledge: Local and wide network communication

Practice Assignment • Grade: 100%

Review: Network architecture

Wrap-up

Video • 39 sec

Glossary terms from module 1

Reading • 4 min

Module challenge 1

Graded Assignment • Grade: 100%

Module 2

Network operations

Introduction to network protocols

Welcome to module 2

Video • 1 min

Network protocols

Video • 3 min

Common network protocols

Reading • 4 min

Additional network protocols

Reading • 4 min

Arturo: Working in network security

Video • 2 min

Wireless protocols

Video • 1 min

The evolution of wireless security protocols

Reading • 4 min

Test your knowledge: Recognize different network protocols

Practice Assignment • Grade: 100%

System identification

Firewalls and network security measures

Video • 4 min

Virtual private networks (VPNs)

Video • 2 min

Security zones

Video • 3 min

Subnetting and CIDR

Reading • 4 min

Proxy servers

Video • 3 min

Virtual networks and privacy

Reading • 4 min

VPN protocols: Wireguard and IPSec

Reading • 4 min

Test your knowledge: System identification

Practice Assignment • 10 min

Review: Network operations

Wrap-up

Video • 51 sec

Glossary terms from module 2

Reading • 4 min

Module challenge 2

Graded Assignment • 50 min

Module 3

Secure against network intrusions

Module 4

Security hardening

Subnetting and CIDR

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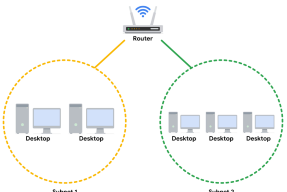
Earlier in this course, you learned about network segmentation, a security technique that divides networks into sections. A private network can be segmented to protect portions of the network from the internet, which is an unsecured global network.

For example, you learned about the uncontrolled zone, the controlled zone, the demilitarized zone, and the restricted zone. Feel free to review the video about [security zones](#) for a refresher on how network segmentation can be used to add a layer of security to your organization's network operations. Creating security zones is one example of a networking strategy called subnetting.

Overview of subnetting

Subnetting is the process of taking one large network and **dividing it into several smaller, organized groups** called **subnets**. Think of it like this: if your company's network is a large city, each subnet is a distinct neighborhood within that city.

Subnetting divides up a network address range into smaller subnets within the network. These organized subnets are defined by the unique combination of the **IP address** and the **network mask** assigned to every device, effectively creating a "network within a network." Subnetting creates a network of devices to function as their own network. This makes the network more efficient and can also be used to create security zones, if devices on the same subnet communicate with each other, the switch changes the transmissions to stay on the same subnet, improving speed and efficiency of the communications.



Classless Inter-Domain Routing notation for subnetting

Classless Inter-Domain Routing (CIDR) is a method of assigning subnet masks to IP addresses to create a subnet. Classless addressing replaces classful addressing. Classful addressing was used in the 1980s as a system of grouping IP addresses into classes (Class A to Class E). Each class included a limited number of IP addresses, which were depleted as the number of devices connecting to the internet outgrew the classful range in the 1990s. Classless CIDR addressing expanded the number of available IPv4 addresses.

CIDR allows cybersecurity professionals to segment classful networks into smaller chunks. CIDR IP addresses are formatted like IPv4 addresses, but they include a slash ("/") followed by a number at the end of the address. This extra number is called the IP network prefix. For example, a regular IPv4 address uses the 198.51.100.0 format, whereas a CIDR IP address would include the IP network prefix at the end of the address, 198.51.100.0/24. This CIDR address encompasses all IP addresses between 198.51.100.0 and 198.51.100.255. The system of CIDR addressing reduces the number of entries in routing tables and provides more available IP addresses within networks. You can try converting CIDR to IPv4 addresses and vice versa through an online conversion tool, like [IPAddressGuide](#) (2), for practice and to better understand this concept.

Note: For now, focus on understanding the concept of CIDR as a flexible addressing method used for modern subnetting. A deeper dive into the technical mathematics of CIDR is a valuable skill you can develop further in future training if needed.

Security benefits of subnetting

Subnetting allows network professionals and analysts to create a network within their own network without requesting another network IP address from their internet service provider. This process uses network bandwidth more efficiently and improves network performance. Subnetting is one component of creating isolated subnetworks through physical isolation, routing configuration, and firewalls.

Key takeaways

Subnetting is a common security strategy used by organizations. Subnetting allows organizations to create smaller networks within their private network. This improves the efficiency of the network and can be used to create security zones.

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Give me a summary

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This reading explains subnetting as a networking strategy that enhances efficiency and security.

Understanding Subnetting

- Subnetting involves dividing a network into smaller, logical groups called subnets, essentially creating a "network inside a network."
- This process improves network efficiency by allowing devices within the same subnet to communicate directly, without involving the main network.

Classless Inter-Domain Routing (CIDR)

- CIDR is a modern method for assigning subnet masks to IP addresses, replacing the older "classful addressing" system.
- CIDR IP addresses include an IP network prefix (e.g., 198.51.100.0/24), which specifies the range of IP addresses within that subnet.

Security Benefits of Subnetting

- Subnetting allows organizations to create isolated subnetworks, enhancing security by segmenting different parts of the network.
- This strategy helps in creating security zones and managing network bandwidth more efficiently without needing additional IP addresses from an internet service provider.

And if you want to continue exploring this topic, try one of these follow-up questions:

What is Classless Inter-Domain Routing (CIDR) in subnetting?

How can you apply subnetting to improve network security?

How does subnetting relate to creating security zones in a network?

Ask me anything

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